

Replication of the virtual experiment by Schmitt *et al.* (2020)

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Using the TROLL simulator in a long-term virtual experiment, Schmitt *et al.* (2020) found that functional diversity improves the resilience of tropical forests. We reproduce a reduced version of this experiment in this vignette to demonstrate how `rcontroll` can help to easily use TROLL to implement virtual experiments. We use the packages `dplyr` & `reshape2` for data processing, `ggplot2` for figures, `FD` for functional divergence indices, and `rcontroll` for simulations.

```
suppressMessages(library(dplyr))
# reshape2
library(ggplot2)
suppressMessages(library(FD))
suppressMessages(library(rcontroll))
```

Simulation experiment

Experimental design

We simulate $N = 10$ communities for each species richness level through $SR \in [5, 10, 20]$ over 600 years, resulting in 30 communities with species randomly sampled in `TROLLv3_species`.

```
N <- 10
SR <- c(5, 10, 20)
data("TROLLv3_species")
communities <- lapply(rep(SR, each = N),
  function(n) sample_n(TROLLv3_species, n))
names(communities) <- paste0("com", 1:length(communities))
```

Note that we selected randomly species with a small variation in species number while Schmitt *et al.* (2020) optimised functional diversity variation in the community while using a larger variation in species number.

Pre-disturbance simulations

Once the communities were defined, we use the `stack` function of `rcontroll` to simulate 30 forests from the bare ground before disturbance. We use the `autoplot` function of `rcontroll` to represent spatial pattern of the forest before disturbance in the `com1` community (Fig. 1).

```
simulations_pre <- stack(name = "pre",
  simulations = names(communities),
  path = "simulations",
  global = generate_parameters(nbiter = 12*600, OUTPUT_extended = 1),
  species = bind_rows(communities, .id = "simulation"),
  climate = TROLLv3_climatedaytime12,
  daily = TROLLv3_daytimevar,
  verbose = FALSE,
  cores = 1)
save(simulations_pre, file = "simulations/simulations_pre.Rdata")

load("simulations/simulations_pre.Rdata")
subsims <- simulations_pre
subsims@forest <- filter(subsims@forest, simulation == "com1")
rcontroll::autoplot(subsims, what = "spatial", variable = "height") +
  viridis::scale_color_viridis()
```

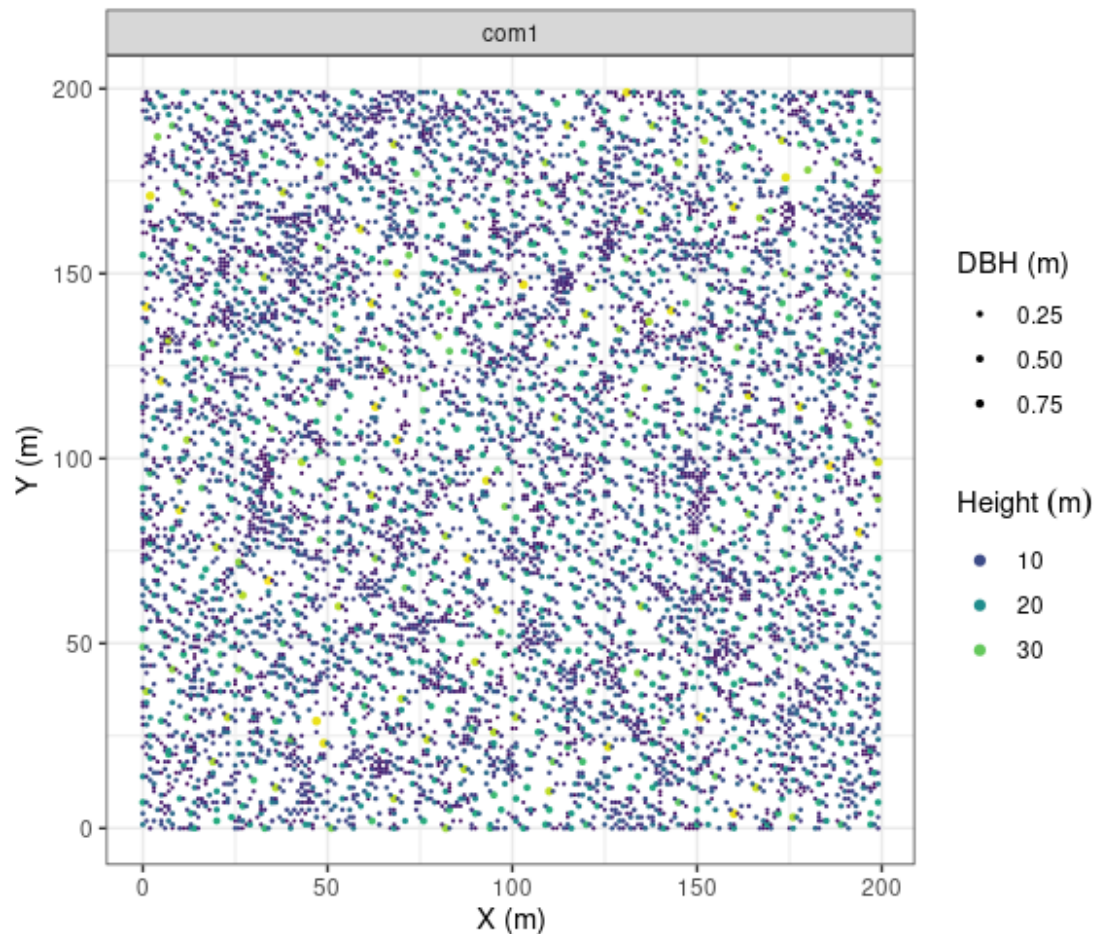


Figure 1: Spatial pattern of the forest before disturbance in the *com1* community. Trees are represented by points with size corresponding to diameter at breast height (DBH, m) and colour to height (m).

Disturbance simulations

Schmitt *et al.* (2020) used a particular type of random disturbance as a first test case to explore the response of forest biodiversity on tropical forest resilience to disturbance. Consequently, we can simply use the `sample_frac` function from `dplyr` to reduce the forest to a random defined fraction per simulation thanks to the `group_by` function from `dplyr`. This results in the spatial pattern of the forest at disturbance in the *com1* community shown in Fig. 2. However, future studies could build on this random disturbance to include more mechanistic details and explore the effect of different disturbance types with homemade R code or specific packages (*e.g.* [LoggingLab](#)).

```

simulations_pre@forest <- simulations_pre@forest %>%
  group_by(simulation) %>%
  sample_frac(1/2)

subsims <- simulations_pre
subsims@forest <- filter(subsims@forest, simulation == "com1")
rcontroll::autoplot(subsims, what = "spatial", variable = "height") +
  viridis::scale_color_viridis()

```

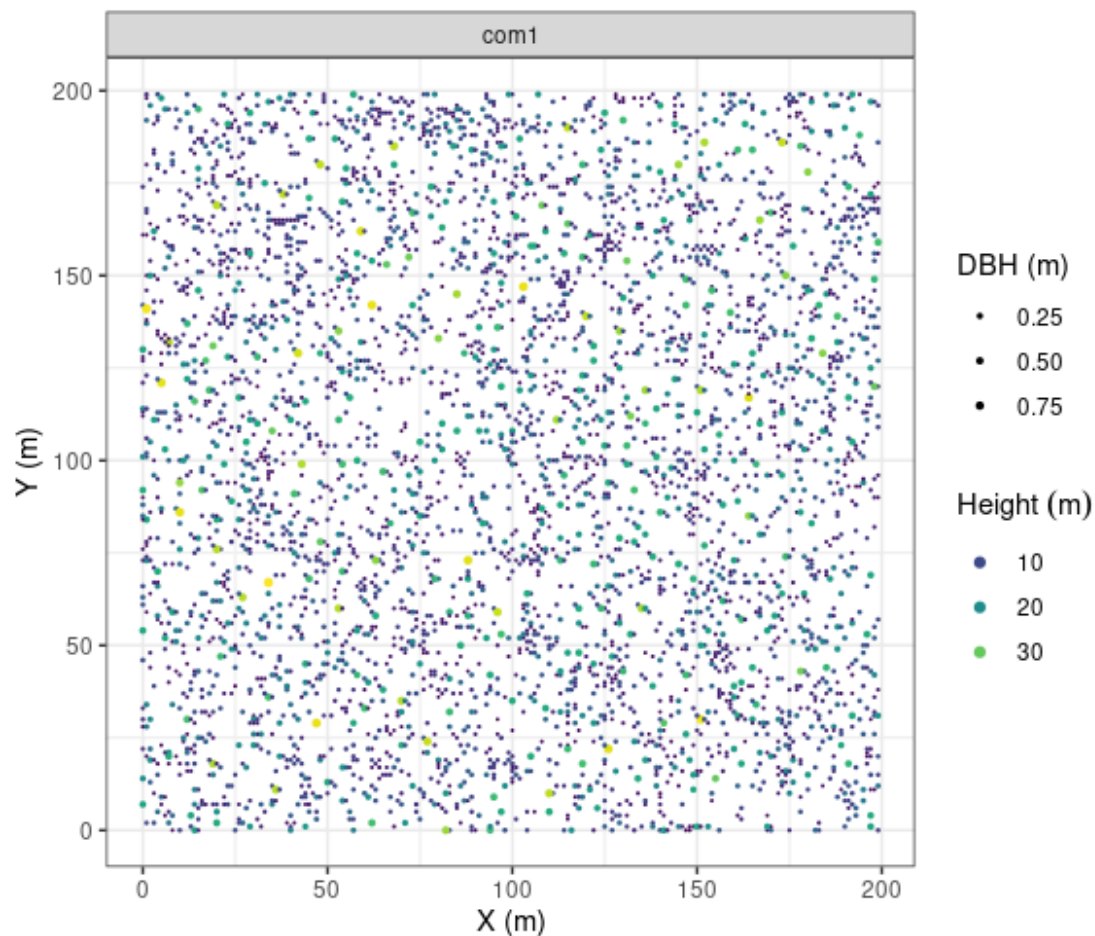


Figure 2: Spatial pattern of the forest after disturbance in the com1 community. Trees are represented by points with size corresponding to diameter at breast height (DBH, m) and colour to height (m).

Note that we removed here a number of stems while Schmitt *et al.* (2020) removed a defined volume of aboveground biomass.

Post-disturbance simulations

Once the communities were disturbed, we use once again the `stack` function of `rcontroll` to simulate the post-disturbance trajectories of the 30 forests without migration (`Cseedrain = 0`). We use again the `autoplot` function of `rcontroll` to represent spatial pattern of the forest after disturbance in the `com1` community (Fig. 3).

```
simulations_post <- stack(name = "post",
  simulations = names(communities),
  path = "simulations",
  global = update_parameters(simulations_pre, Cseedrain = 0),
  species = simulations_pre@inputs$species,
  climate = TROLLv3_climatedaytime12,
  daily = TROLLv3_daytimevar,
  forest = as.data.frame(simulations_pre@forest),
  verbose = FALSE,
  cores = 30)
save(simulations_post, file = "simulations/simulations_post.Rdata")

load("simulations/simulations_post.Rdata")
subsims <- simulations_post
subsims@forest <- filter(subsims@forest, simulation == "com1")
rcontroll::autoplot(subsims, what = "spatial", variable = "height") +
  viridis::scale_color_viridis()
```

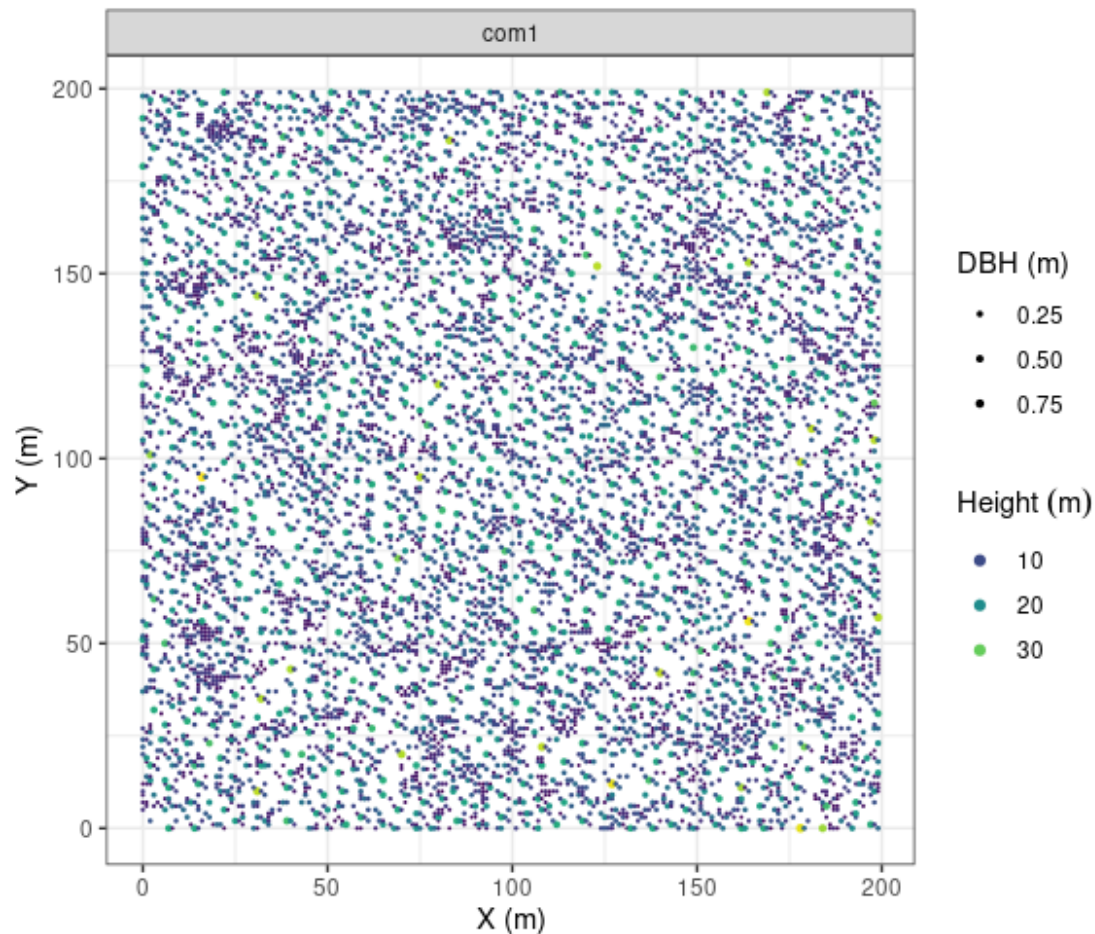



Figure 3: Spatial pattern of the forest 600 years after disturbance in the *com1* community. Trees are represented by points with size corresponding to diameter at breast height (DBH, m) and colour to height (m).

Undisturbed simulations

For control, we use once again the `stack` function of `rcontrol1` to simulate the undisturbed trajectories of the 30 forests without migration (`Cseedrain = 0`). We use again the `autoplot` function of `rcontrol1` to represent spatial pattern of the undisturbed forest in the *com1* community (Fig. 4).

```
load("simulations/simulations_pre.Rdata")
simulations_undist <- stack(name = "undist",
  simulations = names(communities),
  path = "simulations",
  global = update_parameters(simulations_pre, Cseedrain = 0),
```

```
species = simulations_pre@inputs$species,
climate = TROLLv3_climatedaytime12,
daily = TROLLv3_daytimevar,
forest = as.data.frame(simulations_pre@forest),
verbose = FALSE,
cores = 30)
save(simulations_undist, file = "simulations/simulations_undist.Rdata")

load("simulations/simulations_undist.Rdata")
subsims <- simulations_undist
subsims@forest <- filter(subsims@forest, simulation == "com1")
rcontroll::autoplot(subsims, what = "spatial", variable = "height") +
  viridis::scale_color_viridis()
```

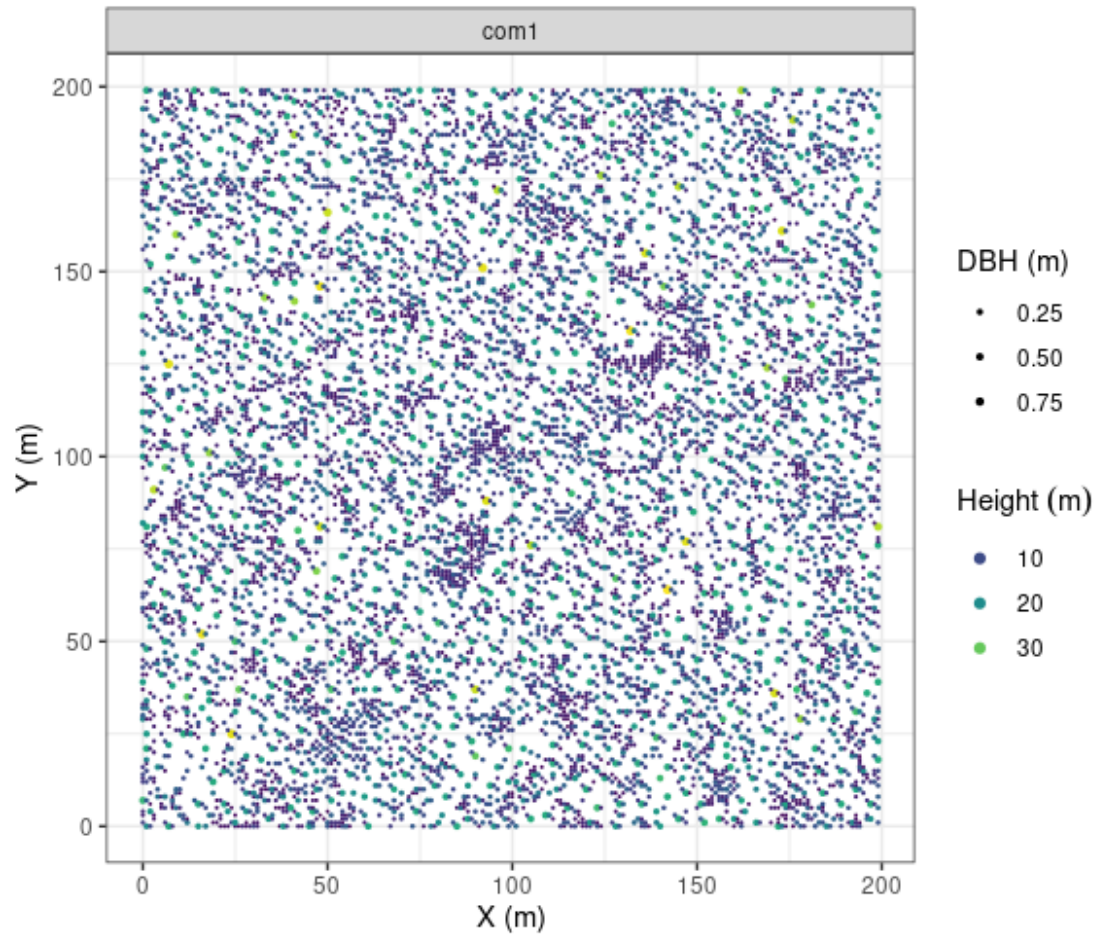


Figure 4: Spatial pattern of the undisturbed forest in the *com1* community. Trees are represented by points with size corresponding to diameter at breast height (DBH, m) and colour to height (m).

Analyses

Quantifying simulated community diversity

Specific and functional diversity are assessed for each simulation just before disturbance, *i.e.* at the end of the pre-disturbance simulation stage, with indices of specific richness (*SR*) and functional diversity (*FDiv*, the volume of the functional space occupied by the community; *FEve*, the regularity of the distribution of abundance in this volume) using the dbFD function from FD. This results in the values shown in Fig. 5.


```

communities_stats <- left_join(simulations_pre@inputs$species,
                              filter(simulations_pre@species, iter ==
max(iter)),
                              by = c("simulation" = "simulation", "s_name" =
"species")) %>%
  group_by(simulation) %>%
  group_by(simulation, s_name) %>% sample_n(1) %>% ungroup() %>%
  mutate(sum1 = ifelse(is.na(sum1), 0, sum1)) %>%
  mutate(abundance = sum1/sum(sum1)) %>%
  select(simulation, s_name, s_LMA, s_Nmass, s_Pmass, s_wsg, s_dbhmax,
s_hmax, s_ah, abundance) %>%
  filter(abundance > 0) %>%
  split(.$simulation) %>%
  lapply(function(tab) {
    tab <- as.data.frame(arrange(tab, s_name))
    row.names(tab) <- tab$s_name
    dbFD(x = tab[c("s_LMA", "s_Nmass", "s_Pmass", "s_wsg", "s_dbhmax",
"s_hmax", "s_ah")],
        a = reshape2::dcast(tab, 1 ~ s_name, value.var = "abundance")[-1], m
= "max", calc.CWM = F)
  }) %>%
  lapply(as.data.frame) %>%
  bind_rows(.id = "community")

ggplot(communities_stats,
        aes(FDiv, FEve, shape = as.factor(nbsp))) +
  geom_point() +
  theme_bw() +
  scale_shape_discrete("Species\nrichness\n(SR)") +
  xlab("Functional diversity (FDiv)") +
  ylab("Functional evenness (FEve)")

```

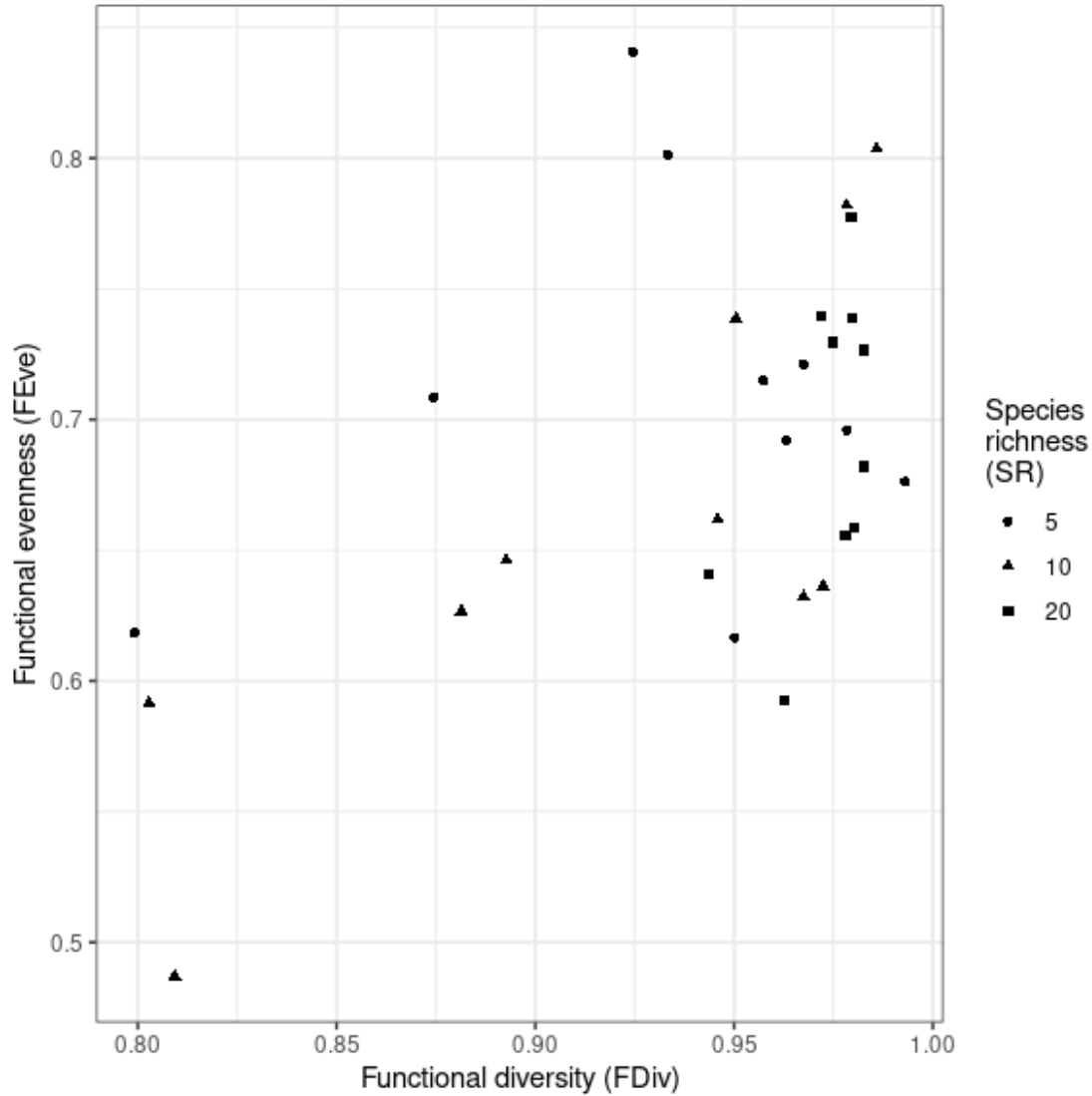


Figure 5: Functional diversity (FDiv) and evenness (FEve) across simulated communities pre-disturbance with varying species richness (SR, shown by point shape).

Note that we have here a smaller variation in functional diversity than Schmitt *et al.* (2020). This is due to the random subsampling of species in the available species, whereas Schmitt *et al.* (2020) optimised the variation.

Ecosystem recovery from disturbance

We characterise changes in ecosystem properties of simulated tropical forests with respect to forest structure (above-ground biomass, AGB in $MgC \cdot ha^{-1}$; basal area, BA in $m^2 \cdot ha^{-1}$; total number of stems, N ; number of stems over 10 cm dbh, N_{10} ; and number of stems over 30 cm dbh, N_{30}). The recovery of the simulated forest at a time t regarding a given property X , $Rx(t)$, is defined relatively to a non-disturbed baseline (Schmitt *et al.* 2020; Fig. 6):

$$R_x(t) = \frac{X_d(t)}{X_c(t)}$$

where $X_d(t)$ and $X_c(t)$ are the values for the ecosystem property X at time t in the disturbed simulation and in the control simulation, respectively. To capture the temporal dimension of resilience, we integrate $d_x(t) = \sqrt{(R_{eq} - R[X(t)])^2}$ through time which quantifies the cumulative difference between disturbed and control runs. We finally compute the inverse of this area as an index of ecosystem resilience, henceforth denoted I_R , so that a higher I_R reflects a higher resilience:

$$I_R(X) = \left(\int_{t=\text{disturbiter}}^{t=\text{disturbiter}+600} d_x(t).dt \right)^{-1}$$

Distance to equilibrium d_{eq} and resilience index I_R are calculated in a multi-dimensional space for forest structure (AGB, BA, N, N10, and N30). The resilience index can be easily computed with the ecosystem attributes of all simulations using dplyr and mathematical formulae (see second chunk below).

```
bind_rows(
  mutate(simulations_pre@ecosystem, state = "pre"),
  mutate(simulations_post@ecosystem, iter = iter + max(iter), state =
"post"),
  mutate(simulations_undist@ecosystem, iter = iter + max(iter), state =
"undist")
) %>%
  filter(simulation %in% sample(unique(.$simulation), 9)) %>%
  ggplot(aes(iter/12, agb, col = state)) +
  geom_line() +
  facet_wrap(~ simulation) +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 90),
    legend.position = "bottom", panel.spacing = unit(0, "lines")) +
  xlab("Time (year)") + ylab("AGB (Kg/ha)")
```

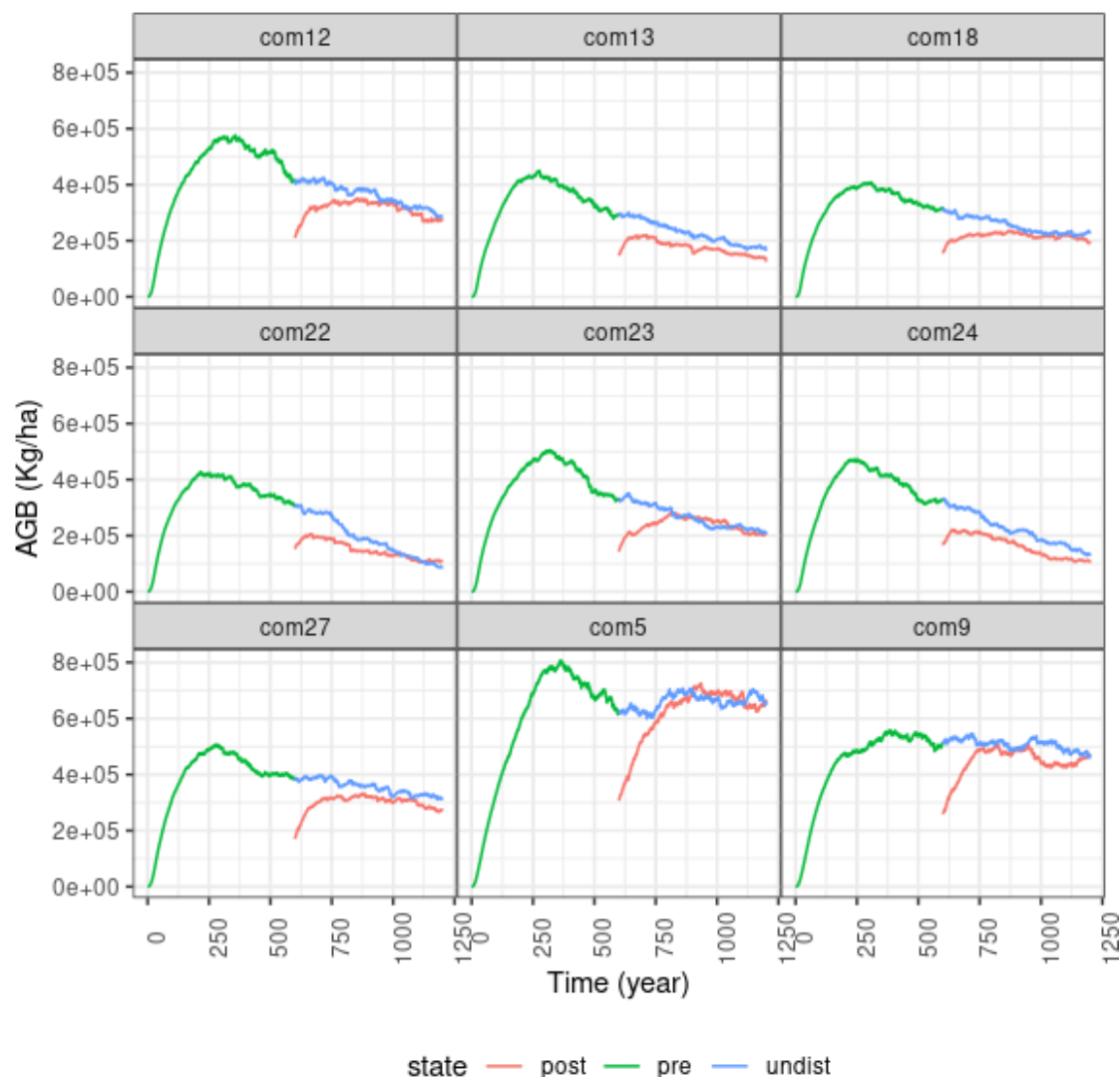


Figure 6: Trajectories of aboveground biomass over time for nine simulated communities, before disturbance (green), after disturbance (blue) and without disturbance (red).

```
resilience <- left_join(
  select(simulations_post@ecosystem, simulation, iter, agb, ba, sum1, sum10,
    sum30) %>%
    reshape2::melt(c("simulation", "iter"), value.name = "X_D"),
  select(simulations_undist@ecosystem, simulation, iter, agb, ba, sum1,
    sum10, sum30) %>%
    reshape2::melt(c("simulation", "iter"), value.name = "X_C")
) %>%
  # filter(iter < 12*200) %>%
  mutate(R = X_D/(X_C+10^-6)) %>%
  group_by(simulation, iter) %>%
  summarise(d_x = sqrt(sum((1-R)^2))) %>%
```

```
group_by(simulation) %>%
summarise(I_R = 1/sum(d_x))
```

Biodiversity effect

We test the influence of three factors on the resilience index I_R : pre-disturbance species richness (SR), functional evenness ($FEve$) and functional diversity ($FDiv$) as shown Fig. 7

```
data <- communities_stats %>%
  left_join(resilience, by = c("community" = "simulation")) %>%
  mutate(SR = nbsp) %>%
  select(community, I_R, SR, FDiv, FEve)

ggplot(data, aes(FDiv, I_R)) +
  geom_point(aes(col = FEve, size = SR)) +
  theme_bw() +
  viridis::scale_color_viridis("Functional\nevenness\n(FEve)") +
  xlab("Functional diversity (FDiv)") +
  ylab("Resilience indice Ir") +
  scale_size_continuous("Species\nrichness\n(SR)")
```

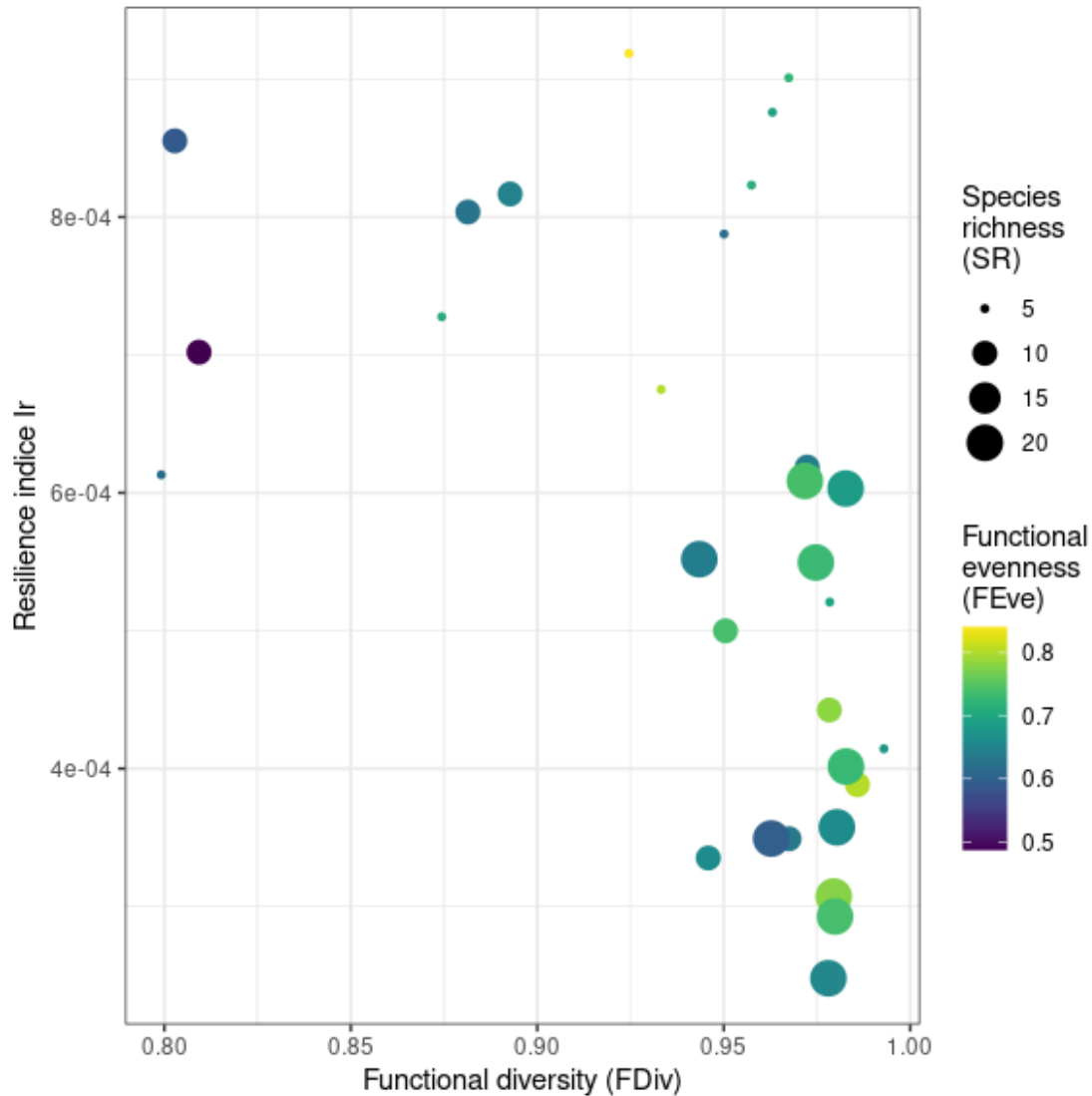


Figure 7: Variation of forest resilience with taxonomic and functional diversity. Resilience index I_R was represented against functional diversity (FDiv); dot size varies with species richness (SR) whereas dot colour indicates functional evenness (FEve).

Conclusion

Using a reduced virtual experiment, we reproduced the analyses from Schmitt *et al.* (2020). This was performed using far less code and time than in the original study using homemade code with an older version of TROLL (S. Schmitt pers. com.).

References

Schmitt, S., Maréchaux, I., Chave, J., Fischer, F.J., Piponiot, C., Traissac, S. & Hérault, B. (2020). Functional diversity improves tropical forest resilience: Insights from a long-term virtual experiment (L. Gomez-Aparicio, Ed.). *Journal of Ecology*, **108**, 831–843. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1111/1365-2745.13320>
<https://onlinelibrary.wiley.com/doi/10.1111/1365-2745.13320>